

EXERCISE NO.3

1. Write a Java program to get the character at the given index within the String.

Sample Output:

Original String = Java Exercises!

The character at position 0 is J

The character at position 10 is i

Code:

```
public class EntryProgram {  
    public static void main(String[] args) {  
        String str = "Java Exercises!";  
        System.out.println("Original String = "+str);  
        char ch1 = str.charAt(0);  
        char ch2 = str.charAt(10);  
        System.out.println("The Character at position 0 is "+ch1);  
        System.out.println("The Character at position 10 is "+ch2);  
    }  
}
```

2. Write a Java program to compare two strings lexicographically. Two strings are lexicographically equal if they are the same length and contain the same number of characters in the same positions.

Sample Output:

String 1: This is Exercise 1

String 2: This is Exercise 2

"This is Exercise 1" is less than "This is Exercise 2"

Code:

```
public class EntryProgram {  
    public static void main(String[] args) {  
        String str1 = "This is Exercise 1";  
        String str2 = "This is Exercise 2";  
        int result = str1.compareTo(str2);  
        if(result < 0)  
            System.out.println("\"" + str1 + "\" is less than \"" + str2 + "\"");  
        else if(result > 0)  
            System.out.println("\"" + str2 + "\" is less than \"" + str1 + "\"");  
        else  
            System.out.println("\"" + str1 + "\" is equal to \"" + str2 + "\"");  
    }  
}
```

3. Write a Java program to check whether a given string ends with the contents of another string.

Sample Output:

"Python Exercises" ends with "se"? false

"Python Exercise" ends with "se"? true

Code:

```
public class EntryProgram {  
    public static void main(String[] args) {  
        String str1 = "Python Exercises";  
        String str2 = "Python Exercise";  
        int result = str1.compareTo(str2);  
        System.out.println "\"" + str1 + "\" ends with \"se\"? " + str1.endsWith("se"));  
        System.out.println "\"" + str2 + "\" ends with \"se\"? " + str2.endsWith("se"));  
    }  
}
```

4. Write a Java program to get the index of all the characters of the alphabet.

Sample Output:

a b c d e f g h i j

=====

36 10 7 40 2 16 42 1 6 20

k l m n o p q r s t

=====

8 35 22 14 12 23 4 11 24 31

u v w x y z

=====

5 27 13 18 38 37

Sample string of all alphabet: "The quick brown fox jumps over the lazy dog."

Code:

```
public class EntryProgram {
    public static void main(String[] args) {
        String str = "The quick brown fox jumps over the lazy dog.";
        for(char ch = 'a'; ch <= 'z'; ch++) {
            for(int i = 0; i<str.length(); i++) {
                if(str.charAt(i) == ch) {
                    System.out.print(ch + " = " + i + " ,");
                    break;
                }
            }
        }
    }
}
```

5. Write a Java program to replace each substring of a given string that matches the given regular expression with the given replacement.

Sample string : "The quick brown fox jumps over the lazy dog."

In the above string replace all the fox with cat.

Sample Output:

Original string: The quick brown fox jumps over the lazy dog.

New String: The quick brown cat jumps over the lazy dog.

Code:

```
public class EntryProgram {
    public static void main(String[] args) {
        String str = "The quick brown fox jumps over the lazy dog.";

        String newStr = str.replaceAll("fox", "cat");
        System.out.println("Original string: " + str);
        System.out.println("New String: " + newStr);
    }
}
```

6. Write a Java program to convert all the characters in a string to uppercase.

Sample Output:

Original String: The Quick BroWn FoX!

String in uppercase: THE QUICK BROWN FOX!

Code:

```
public class EntryProgram {  
    public static void main(String[] args) {  
        String str = "The Quick BroWn FoX!";  
  
        String upper = str.toUpperCase();  
        System.out.println("Original string: " + str);  
        System.out.println("String in uppercase: " + upper);  
    }  
}
```

7. Write a Java program to reverse a string.

Sample Output:

The given string is: The quick brown fox jumps

The string in reverse order is:

spmuj xof nworb kciuq ehT

Code:

```
public class EntryProgram {  
    public static void main(String[] args) {  
        String str = "The quick brown fox jumps";  
        System.out.println("The given string is: " + str);  
        System.out.println("The string in reverse order is:");  
        for (int i = str.length() - 1; i >= 0; i--)  
            System.out.print(str.charAt(i));  
    }  
}
```